

LLM Odyssey

A game-based platform for teaching LLM engineering concepts

Priyamvada Tripathi
Tufts Institute for Artificial Intelligence
Tufts University

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CS graduates get little hands-on LLM engineering practice.

Practitioners name four competencies that programs do not consistently develop.

Prompt engineering

Retrieval-augmented generation

Fine-tuning

Production deployment

Tools are fragmented.

Odyssey is one scaffolded pathway.

What exists

OpenAI Playground

Hugging Face Spaces

Lectures and textbook examples

What Odyssey adds

Progressive difficulty

Pedagogical scaffolding

A path from tokenization to a design brief

Three learning tiers

TIER 1 · GAMES 1-7

COGNITIVE CORE

Remember → Understand → Apply

1 Token Forge

REFERENCE

2 Attention Architect

3 Context Compression

4 Promptsmith

5 Gradient Playground

6 Reasoning Reactor

7 Alignment Arena

Games 2-7 are playable prototypes

TIER 2 · GAMES 8-12

SYSTEMS FORGE

Apply → Analyze → Evaluate

8 Ship-It 9 Agent Architect 10 Retrieval Lab

11 System Composer 12 ProdOps Gauntlet prototypes

TIER 3 · GAME 13

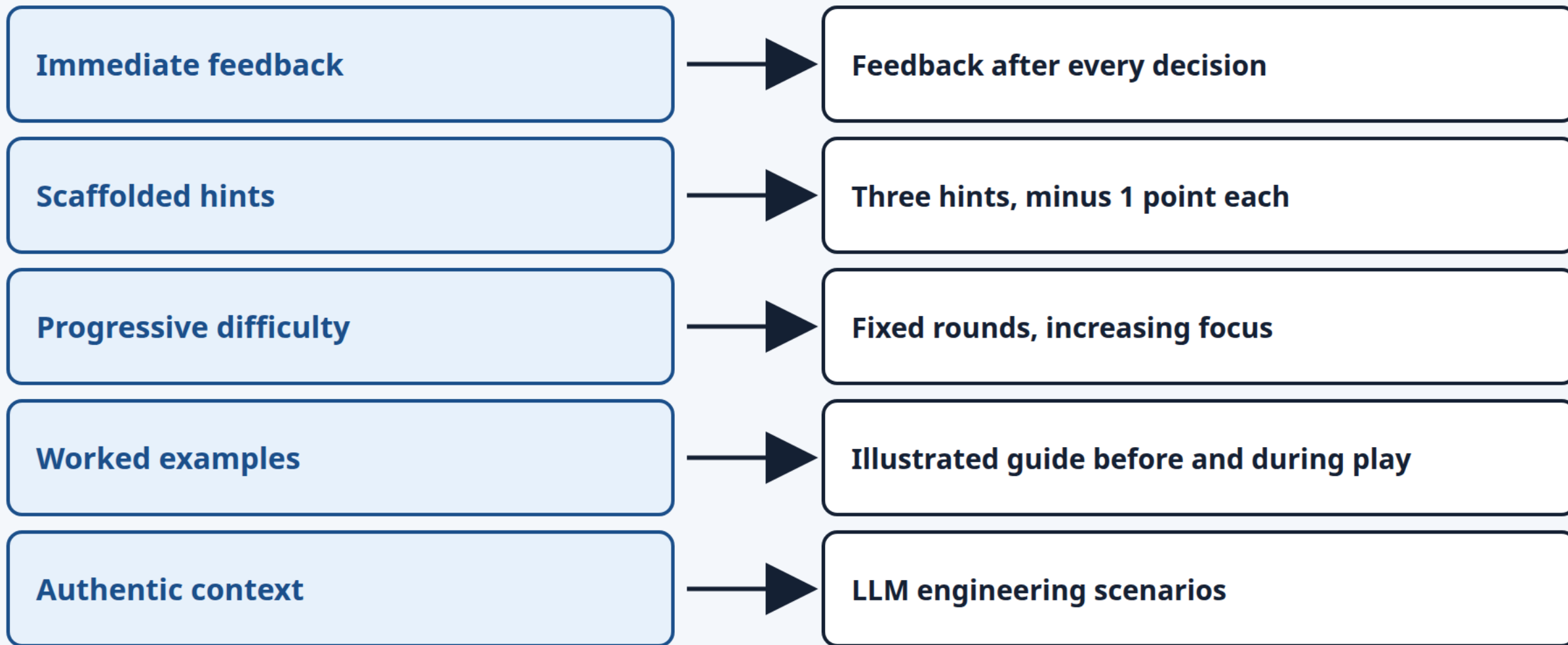
FOUNDRY ARENA

Analyze → Evaluate → Create

Cross-domain capstone · playable prototype

Foundations → Systems → Synthesis

Learning science → game mechanic



Black & Wiliam · Vygotsky · Sweller · Bloom. Practice support, not a measured gain.

```
_def | _refund_total | ( | items | : | _list | [str] | ) | _→ | _int:
```

Tokens
10

Vocabulary efficiency
86%

Cost for 250,000 calls
\$37.50

```
_def _refund_total ( items : _list [str] ) _→ _int:
```

1 Scenario pieces**2 Count and cost**

Preview only. Locking the round runs the scorer. This round is locked.

Hints

Hints used 0/3. Each hint subtracts 1 point from this round after scoring, to a floor of 0.

Feedback

Your decision. SentencePiece (10 authored tokens)

Result. Correct

Why. Keeping the Identifier intact reduced the authored token count for this signature. Code is a different distribution from prose. This comparison was written so SentencePiece keeps the identifier intact. That is a teaching choice, not a benchmark of the SentencePiece library.

What mattered in this scenario? A split identifier is harder to align with the code the caller meant, and it is longer.

Why the alternatives were weaker. Rows that slice the name into many pieces can look thorough and still fail the goal of keeping the Identifier intact.

Trade-off. Splitting snake_case and punctuation makes the prompt longer and harder to align with the code. A compact row can keep the name intact.

Engineering takeaway. Code is a different distribution from prose, so the preferred segmentation can change.

What would happen in a real system? Code-review and tool-calling prompts often pay their token cost in identifiers and punctuation, not in English words.

- SentencePiece is scored as best for this precomputed exercise.
- Token counts — SentencePiece: 10 tokens; Unigram: 11 tokens; BPE: 15 tokens; WordPiece: 16 tokens.
- Estimated cost for 250,000 calls: \$37.50.

Raw points 10. Hint penalty 0. Round score 10 / 10.

Retry this round

Next round

3 Learner choice**4 Immediate feedback**

How one game teaches

1 ORIENTATION

What, why, and how to play.

2 CHALLENGE

An engineering decision.

3 FEEDBACK

Result, why, and the trade-off.

HINT OR CONTINUE

Hint: minus 1. Retry keeps the best.

4 NEXT ROUND

Same pattern until the last round.

CONCEPT GUIDE

Reopen any time.

Round and score stay.

5 MASTERY CHECK

70% benchmark. Not expertise.

6 REFLECT, THEN NEXT

Self-eval does not change the score.

Completion is finishing. Mastery is the benchmark. Transfer is a new case.

Game list

Cognitive Core

1. Token Forge · Reference

2. Attention Architect ·
Prototype3. Context Compression ·
Prototype

4. Promptsmith · Prototype

5. Gradient Playground ·
Prototype6. Reasoning Reactor ·
Prototype

7. Alignment Arena · Prototype

Systems Forge

8. Ship-It Simulator · Prototype

9. Agent Architect · Prototype

10. Retrieval Lab · Prototype

11. System Composer ·
Prototype12. ProdOps Gauntlet ·
Prototype

Foundry Arena

13. Foundry Arena · Prototype

Games, status, and time

#	Game	Tier	Students learn	Status	Minutes	Bloom	Mastery
1	Token Forge	Cognitive Core	See how text becomes the discrete units processed by a language model. Compare subword segmentation strategies and reason about vocabulary, sequence length, multilingual text, code, context use, and token-related cost.	Reference	20	remember, understand, apply	70%
2	Attention Architect	Cognitive Core	Compute a tiny self-attention pattern and read which token receives the weight.	Prototype	25	understand, apply	70%
3	Context Compression	Cognitive Core	Fit source text into a token budget without dropping the fact the task needs.	Prototype	20	understand, apply	70%
4	Promptsmith	Cognitive Core	Match a prompting strategy to a task and read the quality, token, and failure tradeoff.	Prototype	25	apply, analyze	70%
5	Gradient Playground	Cognitive Core	Tune learning rate, epochs, batch size, and full fine-tuning versus LoRA on an educational loss model.	Prototype	25	understand, apply, analyze	70%
6	Reasoning Reactor	Cognitive Core	Assemble a visible reasoning workflow: decomposition, checks, samples, and temperature.	Prototype	25	apply, analyze	70%
7	Alignment Arena	Cognitive Core	Rank replies with an explicit rubric for helpfulness, safety, and factuality.	Prototype	25	understand, analyze, evaluate	70%
8	Ship-It Simulator	Systems Forge	Meet a latency, cost, and reliability target by combining serving levers.	Prototype	25	apply, analyze, evaluate	70%
9	Agent Architect	Systems Forge	Design an agent loop with tools, permissions, checks, and a way to stop.	Prototype	25	apply, analyze, evaluate	70%
10	Retrieval Lab	Systems Forge	Compare keyword, vector, and hybrid retrieval on a tiny local corpus, including a metadata filter.	Prototype	30	apply, analyze, evaluate	70%
11	System Composer	Systems Forge	Assemble a small LLM system that covers grounding, safety, latency, and operations together.	Prototype	25	analyze, evaluate	70%
12	ProdOps Gauntlet	Systems Forge	Respond to cost spikes, regressions, drift, and audit questions with an operational action.	Prototype	25	analyze, evaluate	70%
13	Foundry Arena	Foundry Arena	Design a constrained LLM system on an industry, healthcare, robotics, ethics, education, or sandbox brief.	Prototype	40	analyze, evaluate, create	70%

Learning objectives

1. Token Forge Reference

Make tokenization visible so learners can connect segmentation choices to sequence length and API cost.

Evaluation flow · documented, not proven

PRE-ASSESSMENT

Optional. Local. 10 items.



ODYSSEY USE

The 13 games.



LOCAL LOGS

Time, completion, hints, retries.



POST-ASSESSMENT

Optional. Local. Same instrument.

NOT IN THIS APP

STUDY SURVEY

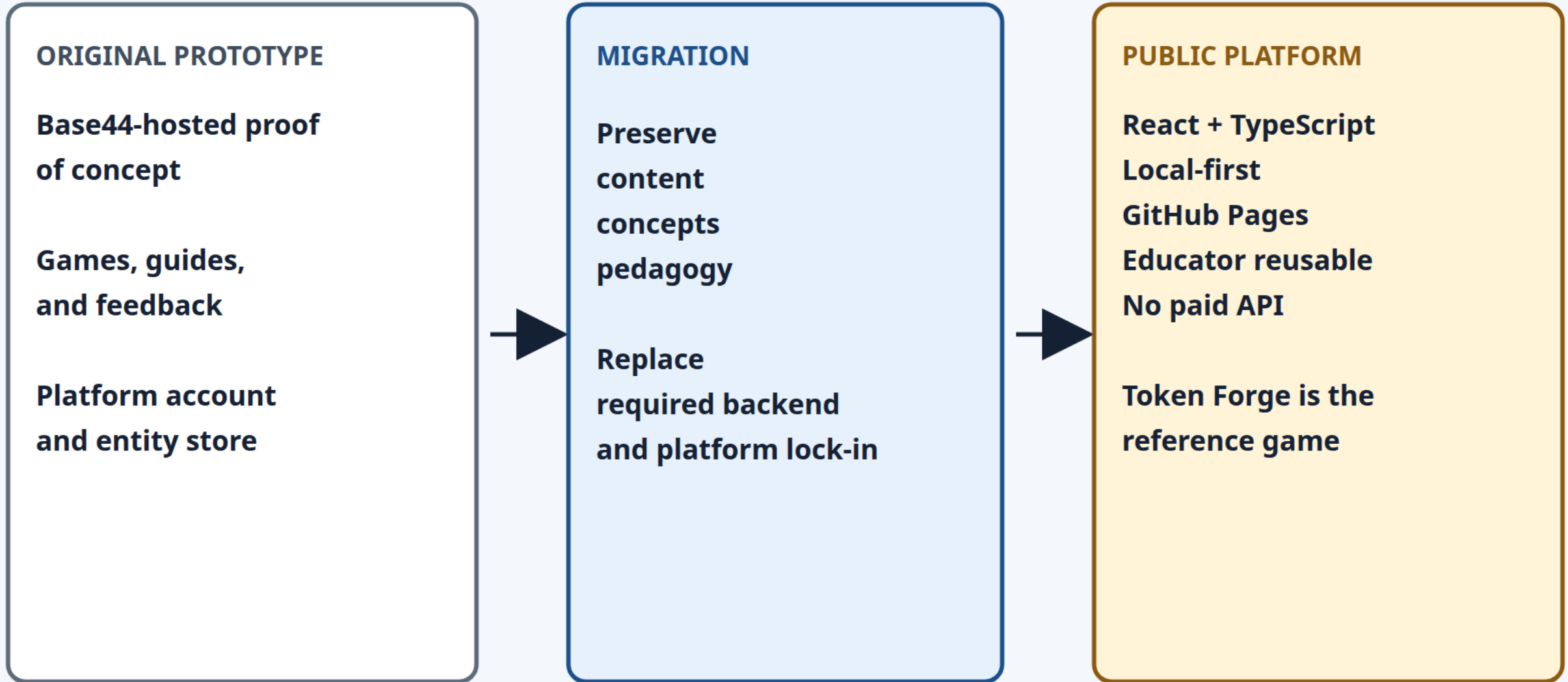
Removed with Base44.
Feedback page is local only.

INTERVIEWS

PLANNED
Not shipped.

Questions for a future study: knowledge, engagement, feature use, prior expertise.

From prototype to public platform



<https://piaattufts.github.io/llmodysey/>

<https://github.com/piaattufts/llmodysey>

Backup • Reported deployment targets

< 16 ms

interaction response

< 100 ms

page load

60 fps

animations

0

availability incidents

Game logic runs in the browser, so play does not wait on a server round trip.

From the work-in-progress deployment report. Not a new benchmark of this public release, and not a learning result.

Backup • Homepage and a Token Forge round

LLM Odyssey

13 games · 3 learning tiers · Browser-based · No paid API required · Open source · Educator reusable

Learn

AI Games Arcade

Concept Index

Progress

Apply

The Foundry

Assess

Pre-assessment

Post-assessment

Feedback

Teach

Educator Guide

Demo

Demo Mode

Your journey

0 of 13 games at the mastery threshold

Current game: Token Forge

Open the progress record

Game list

Cognitive Core

1. Token Forge · Reference

2. Attention Architect · Prototype

3. Context Compression · Prototype

The Odyssey · Master the Machine

LLM Odyssey

Learn LLM Engineering Through Interactive Games

LLM Odyssey is an open-source, browser-based learning environment that turns core LLM engineering concepts into interactive decision-making activities.

Students move from foundational topics such as tokenization and attention to retrieval, agents, production operations, and system design.

13 interactive games

3 learning tiers

No paid API required

Open source

Local-first

Educator reusable

Token Forge is the reference implementation. The other twelve games are playable prototypes. Their scores are calculated, and each card shows that status. Nothing here calls a paid model API.

Start Learning

Explore as Educator

Take the Demo Tour

Continue: Token Forge

[View on GitHub](#)

0 of 13 games mastered. Recommended next: Token Forge.

[Optional pre-assessment](#)

☐ Practice ahead. Opens later tiers before the mastery counts are met. Listed prerequisites stay visible either way.

How the Odyssey works

1. Understand

Read why the game exists, what you will do, and what the screen is actually calculating.

2. Experiment

Make a decision on a scenario. The examples stay the same for every learner.

3. Get feedback

See why the choice fit, why the alternatives were weaker, and what a real system would require.

4. Reflect

Answer a short prompt. The reflection does not change the score.

5. Apply

Use the concept guide and, later, a design brief in Foundry Arena.

6. Progress

Completion, the score, and the mastery threshold are recorded separately. Transfer is whether you can explain a new case.

Threshold

Current game: Token Forge

Open the progress record

Game list

Cognitive Core

1. Token Forge · Reference

2. Attention Architect · Prototype

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5. Gradient Playground · Prototype

6. Reasoning Reactor · Prototype

7. Alignment Arena · Prototype

Systems Forge

8. Ship-It Simulator · Prototype

9. Agent Architect · Prototype

10. Retrieval Lab · Prototype

11. System Composer · Prototype

12. ProxOps Gauntlet · Prototype

Foundry Arena

13. Foundry Arena · Prototype

How this round is scored

Minimize tokens while keeping the function name recognizable.

def refund_total(items: list[str]) → int:

BPE

SELECT

Pairs have not fully merged the snake-case name in this row.

WordPiece

SELECT

Continuation marks appear inside the name and the brackets.

SentencePiece

SELECTED

The identifier and the annotation stay in fewer authored pieces.

Unigram

SELECT

Closer than BPE, but the arrow and brackets are still separate pieces.

_def | _refund_total | (| items | : | _list | [str] |) | _→ | _int:

Tokens

10

Vocabulary efficiency

86%

Cost for 250,000 calls

\$3750

_def | _refund_total | (| items | : | _list | [str] |) | _→ | _int:

Preview only. Locking the round runs the scorer. You can still change the selection.

Hints

Hints used 0/3. Each hint subtracts 1 point from this round after scoring, to a floor of 0.

Reveal next hint

Lock in this round

LLM Odyssey

Homepage

Reference game, before feedback

Backup • A prototype game and the capstone

0 of 13 games at the mastery threshold
Current game: Token Forge
[Open the progress record](#)

Game list

Cognitive Core

1. Token Forge - Reference

2. Attention Architect - Prototype

3. Context Compression - Prototype

4. Promptsmith - Prototype

5. Gradient Playground - Prototype

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Systems Forge

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10. Retrieval Lab - Prototype

11. System Composer - Prototype

12. ProdOps Gauntlet - Prototype

Foundry Arena

13. Foundry Arena - Prototype

Task. Retrieve the refund rule. Pure vector search prefers a distractor.

► How this round is scored

Choose a method. A filter is not available on this corpus.

Query: What is the refund window?

Keyword

Vector - Selected

Hybrid

Price match (1.00)

Precision
0% - not met

Recall
0% - not met

Top hit
Price match

In top-k. Price match - score 1.00 - not relevant.
Price match applies within 14 days and is not a refund.

Below top-k. Refund policy - score 0.22 - relevant to this question
The refund window is 30 days after delivery.

Below top-k. Shipping - score 0.00 - not relevant
Standard shipping takes five business days.

Preview only. Locking the round runs the scorer. You can still change the selection.

Hints
Hints used 0/3. Each hint subtracts 1 point from this round after scoring, to a floor of 0.

Reveal next hint

Lock in this round

LLMOdyssey

Created by Priyanshu Tripathi
Tufts Institute for Artificial Intelligence
Tufts University

[Project](#) [GitHub](#) [Citation](#) [Research paper](#) [Contact](#)

Retrieval Lab, playable prototype

0 of 13 games at the mastery threshold
Current game: Token Forge
[Open the progress record](#)

Game list

Cognitive Core

1. Token Forge - Reference

2. Attention Architect - Prototype

3. Context Compression - Prototype

4. Promptsmith - Prototype

5. Gradient Playground - Prototype

6. Reasoning Reactor - Prototype

7. Alignment Arena - Prototype

Systems Forge

8. Ship-It Simulator - Prototype

9. Agent Architect - Prototype

10. Retrieval Lab - Prototype

11. System Composer - Prototype

12. ProdOps Gauntlet - Prototype

Foundry Arena

13. Foundry Arena - Prototype

Task. Cover cost, grounding, and escalation for a customer-support assistant.

► How this round is scored

Choose one option in each decision, then reflect.

Predictable model spend

Return answers quote the manual

Refunds above \$200 reach a person

How do ordinary questions get a model?

Route repeats to a cache and a small model
Keeps spend predictable.

SELECT

Send every call to the largest model
Spend tracks traffic with no cap.

SELECT

Where do return answers come from?

Retrieve the return manual and cite the section
The answer can be checked.

SELECT

Let the model recall store policy from training
No section to audit.

SELECT

What happens above \$200?

Open a ticket for a person
The model stops.

SELECT

Let the model issue the refund
No person in the path.

SELECT

Reflection

Self-assessment

Spend stays predictable

1

2

3

4

Foundry Arena, playable prototype

From prototype to public platform

ORIGINAL PROTOTYPE

Base44-hosted proof
of concept

Games, guides,
and feedback

Platform account
and entity store



MIGRATION

Preserve
content
concepts
pedagogy

Replace
required backend
and platform lock-in



PUBLIC PLATFORM

React + TypeScript
Local-first
GitHub Pages
Educator reusable
No paid API

Token Forge is the
reference game

Product evolution. The prototype supplied the curriculum. This release stands alone.

Backup • Feasibility feedback, not a learning result

Two faculty reviewers, Winter 2026.

The author was then at Durham College. That is not the current affiliation.

They confirmed the platform runs.

They named adaptive difficulty as the next priority.

This is face-validity feedback. It is not evidence of learning effectiveness.